

AphanesKyma

Hearing an image, and hiding one – two frequency-domain tricks

Samia Azrin Aunkon Dhriti Roy Orchi
Discrete Duo

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Motivation & problem

Part I – an audio channel for images

Can a picture be carried through the air on ordinary sound – speaker to microphone – and rebuilt despite noise, echo, and clock mismatch?

Part II – invisible, secure payloads

Can we hide a message or image *inside* an ordinary cover photo so it survives a PNG re-save, and only the password holder can recover it?

One question underneath both

Where in a transform domain can information be placed so it survives a lossy, noisy, real-world round trip? Part I answers this for a *temporal* channel (air); Part II for a *spatial* one (image rounding). Two independent pipelines, kept separate below, share that one idea.

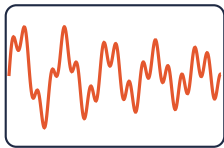
PART I

Turning an image into sound, and back

Have we ever wondered what an image would sound like?



a picture



a sound wave



the picture again, after air

Goal: play an image through a speaker, record it with a microphone, and rebuild it.

Core idea: a spectrogram is already an image

Spectrogram: **time** $\times$ **frequency**, brightness = magnitude. We run it backwards.

Mapping, for an image $I \in \mathbb{R}^{H \times W}$

row $i \rightarrow$ time frame i

column $k \rightarrow$ frequency bin k

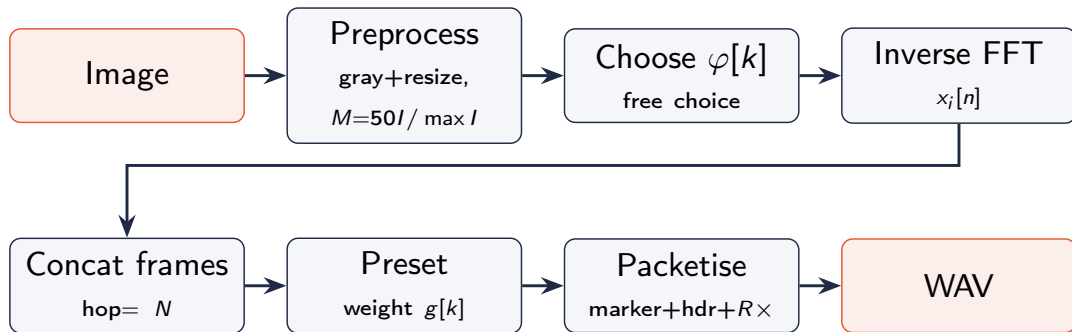
pixel $I[i, k] \rightarrow$ magnitude $|X_i[k]|$

$$f_k = \frac{k f_s}{N}, \quad N = 2(W - 1), \quad f_s = 48 \text{ kHz}$$

$$T = \frac{H N}{f_s} \approx 2.7 \text{ s for } 256 \times 256$$

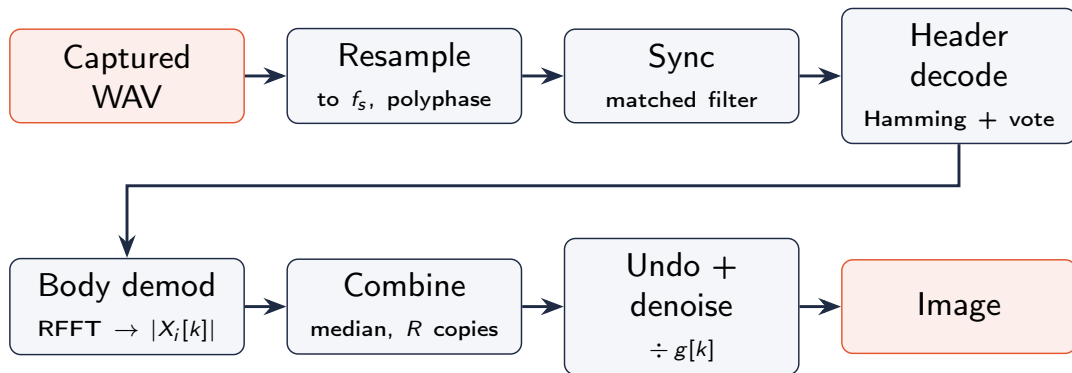


Sonification pipeline: encoding, in detail



Phase never carries information – only the magnitude $M[i, k]$ does, so the decoder can discard $\varphi[k]$ entirely.

Sonification pipeline: decoding, in detail



Median combining and Hamming voting exploit the same idea: redundancy survives a fading channel, a single bad copy doesn't.

Encoder: image to audio

Preprocess: grayscale $I = 0.299R + 0.587G + 0.114B$, Lanczos resize to ≤ 256 , scale $M = 50 I / \max I$.

Per row, choose a phase $\varphi[k]$ and apply the inverse real FFT:

$$X_i[k] = M[i, k] e^{j\varphi[k]}, \quad x_i[n] = \frac{1}{N} \sum_k X_i[k] e^{j2\pi kn/N}$$

Frames are concatenated (rectangular window) and peak-normalised.

Detail that matters

The inverse real FFT drops the imaginary part of bins 0 and $N/2$, so we pin their phase to 0 to keep those columns exact.

Phase is a free choice: random (hiss), zero (pulses), chirp $\varphi = ck^2/W$, min-phase, borrowed.

The decoder takes $|\cdot|$, so every phase mode recovers the same image.

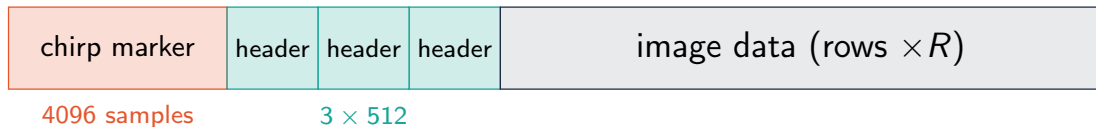
Making it kinder to the ear: presets

Hearing peaks near 2–5 kHz. The *balanced* preset attenuates those bins with the A-weighting curve, $s[k] = 10^{A(f_k)/20}$:

$$g[k] = (1 - \text{floor}) \frac{s[k]^{-\alpha}}{\max_k s[k]^{-\alpha}} + \text{floor}, \quad \tilde{M} = g \odot (M * \text{box})$$

Preset	α	floor	Row blur
fidelity	0	1.0	none (best image quality)
balanced	0.6	0.1	5% (quieter, lossy)

What is played, and how the receiver finds it



Marker: linear chirp $500 \rightarrow 18\,000$ Hz. The receiver correlates the recording with the clean template:

$$\rho[n] = \frac{\langle x[n:n+L], m \rangle}{\|x[n:n+L]\| \|m\|}, \quad \hat{n} = \arg \max \rho$$

Accept if $\rho_{\max} \geq 0.45$ (retry at 0.35). A sweep has a sharp autocorrelation peak, so it survives reverb.

Header (40 bits): rows, cols, preset, repeats.

- Hamming(7,4): $s = Hr$ gives the error position
- bit interleaving against burst errors
- one always-on **pilot bin** per chunk gives a local threshold
- 3 copies, majority vote

Surviving the real world

Sample-rate mismatch

The mic records at its own rate (read from the WAV header).

Resample by $U/D = f_s/f_{\text{native}}$ with a polyphase filter, or frames misalign.

Unknown start

Matched filter finds the marker; then search ± 32 samples and keep the offset with a *valid* header and the largest pilot energy.

Row repeats

Each row is sent R times (≤ 15 , stored in the header). Decode with a per-bin median:

$$\hat{l}[i, k] = \text{median}_r |X_{i,r}[k]|$$

The median ignores corrupted copies (a mean would be dragged by one outlier), at the price of $R \times$ longer audio.

Decoder and denoising

$$\tilde{I}[i, k] = |\text{RFFT}\{x[iN : (i+1)N]\}[k]| \rightarrow \text{median over } R \rightarrow \tilde{I}/g[k] \rightarrow \text{denoise} \rightarrow \text{PNG}$$

Denoiser	Operation	Best for
median 3×3	local median	AWGN speckle, clipping
box blur	separable average	dropout bands
spectral	2D DFT, keep central 35%	broadband noise
none	pass-through	band-limited channel

No single filter wins: the best one depends on how the channel corrupts the signal.

Results: image $\rightarrow$ audio $\rightarrow$ air $\rightarrow$ image



original



clean round trip



speaker $\rightarrow$ mic, raw



speaker $\rightarrow$ mic, 2D low
pass

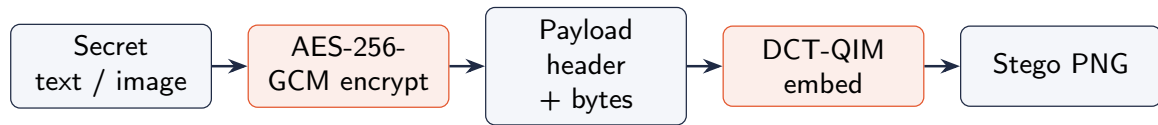
Clean: exact up to a global scale ($\text{RFFT} \circ \text{IRFFT} = \text{id}$). Through air: reverb and noise degrade it, but marker, protected header, repeats and denoising keep the picture.

PART II

Hiding a secret inside an ordinary-looking image

Part II: hiding a secret in plain sight

A second pipeline: put an encrypted message *or image* inside an ordinary-looking cover image.



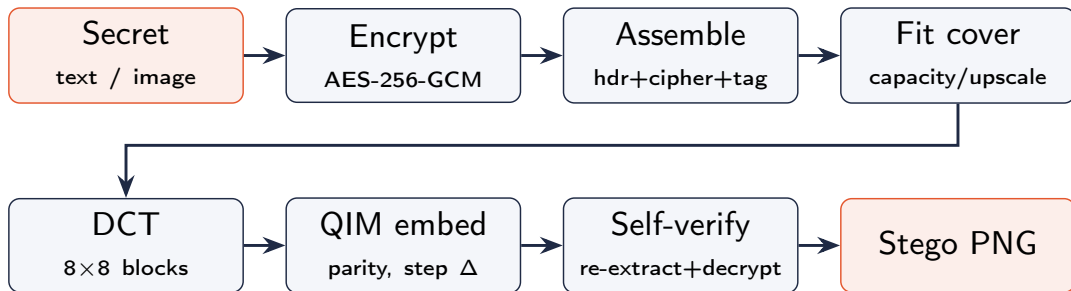
Two layers of protection

- *cryptography*: nobody can read it without the password
- *steganography*: nobody knows it is there

Why not plain LSB?

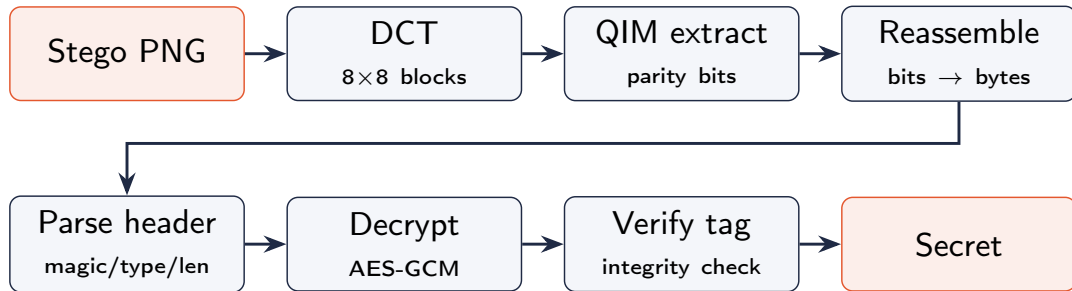
The first prototype flipped least-significant bits of RGB (3 bits/pixel). Simple, but any re-quantisation destroys it. We embed in the DCT domain instead.

Encryption pipeline, in detail



Self-verification re-runs decryption on the saved file before it ever leaves the server – a broken embed is caught, not shipped.

Decryption pipeline, in detail



A wrong password or any tampering fails the tag check here – not silently.

Cryptography and payload format

AES-256-GCM (authenticated encryption)

$$K = \text{SHA256}(\text{password} \parallel \text{salt})$$
$$(C, \tau) = \text{AES-GCM}_K(\text{nonce}, D)$$

Output: salt(16) | nonce(12) | ciphertext | tag(16)

The tag gives **integrity**: a wrong password or a flipped bit makes decryption fail.

Payload header (10 bytes)

APKY	ver	type	length
4 B	1 B	1 B	4 B

type = 1 text, 2 image. Fixed overhead:
 $10 + 12 + 16 + 16 = 54$ bytes.

Honest limitation

A single SHA-256 is fast, so weak passwords can be guessed quickly. A slow KDF (scrypt, Argon2) is the natural upgrade.

Embedding: quantisation index modulation in the DCT domain

Split the cover (grayscale) into 8×8 blocks B ; take the orthonormal 2D DCT:

$$C = D B D^T$$

Hide **8 bits per block** in the mid-frequency coefficients

$(1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)$: low enough to be robust, away from the DC term.

QIM (step $\Delta = 16$): the *parity* of the quantisation index carries the bit.

$$i = \left\lfloor \frac{|c|}{\Delta} \right\rfloor, \quad c' = \text{sgn}(c) \Delta \tilde{i}, \quad \tilde{i} \equiv b \pmod{2} \text{ (nearest such index)}$$

$$\text{extract: } \hat{b} = \left\lfloor \frac{|c'|}{\Delta} \right\rfloor \bmod 2$$

Any noise smaller than $\Delta/2$ leaves $\hat{b} = b$, so the bits survive uint8 rounding. An FFT-magnitude variant with conjugate symmetry is also implemented.

Capacity

$$\left\lfloor \frac{H}{8} \right\rfloor \left\lfloor \frac{W}{8} \right\rfloor \cdot 8 \text{ bits} = \text{one byte per block}$$

If the payload does not fit, the cover is upscaled (Lanczos, same aspect ratio) with a $1.25\times$ margin, so the payload fills $\leq 80\%$ of capacity.

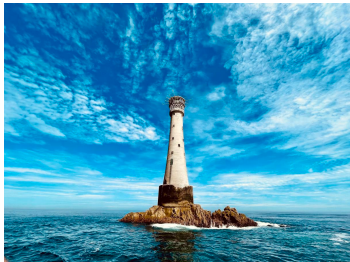
Failure guards

- pixels compressed $[0, 255] \rightarrow [16, 239]$ before embedding, so the IDCT does not clip
- **self-verification**: the server re-extracts and decrypts the saved file before returning it

Quality: $\text{PSNR} = 10 \log_{10} \frac{255^2}{\text{MSE}}$

Decoding and results

Decode: read 10-byte header → check magic, version, type → read length → extract payload
→ AES-GCM decrypt → text or image.



cover



stego (PSNR: ... dB)



recovered secret

Take-away

- Two independent pipelines, one shared idea: pick the right place in a transform domain to put information so it survives a lossy round trip.
- **Part I:** DFT *magnitude* is the message, *phase* a free design knob; synchronisation (chirp + matched filter) and layered redundancy (Hamming, repeats, median) make it work through air.
- **Part II:** authenticated encryption plus DCT-domain QIM hides data robustly, and self-verification catches failures early.
- **Common thread:** the Fourier family of transforms (DFT, DCT) is the working tool in both halves.

Next: slow KDF, colour images, adaptive R , and sending a stego image through the acoustic channel.

Questions?